2026 SHENZHEN FINTECHATHON / TOPIC C

CEKBAO

Student Payment Verification Agent

Before tuition payments, cross-check the request against independently sourced university procedures.

HMIF | INDONESIA | PRE-SUBMISSION DRAFT

Real fraud, measurable failure modes

SINGAPORE | MARCH 2026

20+ reported cases; S$31,000+ in losses from education-institution impersonation. Urgent tuition emails sent victims toward external transfers.

INDONESIA | UMTS 2025

About 273 students reportedly affected. 28 slips did not match the six bank entries investigated. Initial discrepancy ~IDR 1.2bn; later audit figure differed.

Sources: Singapore Police Force (19 Mar 2026); detikSumut (23 Feb 2025; 26 Jan 2026). No national fraud rate inferred.

A student needs source-backed verification

CURRENT: PROVIDERS

Flywire supports education payments. A payment rail does not prove a random external email is authorized.

CURRENT: SCAM TOOLS

ScamShield checks suspicious communications. Generic classification may lack a campus policy reconciliation workflow.

HYPOTHESIS: CEKBAO

Cross-check specific campus policy, receipt evidence and next steps, with citations and explicit uncertainty.

USP is a testable workflow hypothesis, NOT a claim that competitors lack every function.

One controlled pre-payment workflow

01

EXTRACT

Request facts

>

02

COMPARE

Campus policy

>

03

RECONCILE

Synthetic ledger

>

04

ESCALATE

Official channel

Current: deterministic policy checks. Pending: authenticated AI extraction, protected policy sources and auditable user approval.

Offline baseline has reproducible evidence

4 / 4 CASES

Spoofed payee and domain; unconfirmed synthetic tuition slip; matching synthetic policy; unknown institution.

10 / 10 TESTS

All ten local unit tests passed. This verifies only the deterministic checker and synthetic fixture scenarios.

Not tested: live LLM inference, bank confirmation, institutional integrations or user effectiveness.

Safety must be enforceable, not asserted

BASELINE CHECKS

Exact HTTPS host comparison; payee/channel mismatch; unknown institution refusal; synthetic ledger conflict; no settlement claim.

GAPS TO CLOSE

Server authentication; object ownership; prompt injection; immutable audit; data retention; access tiers; demo with real model.

The current sandbox does not meet the required permission-tier implementation standard.

Example 1: prevent unsafe financial action

INPUT / Unverified university lookalike link + third-party payee

RESULT / REQUIRES_OFFICIAL_CHECK

Flags: REQUEST_DOMAIN_MISMATCH, PAYEE_MISMATCH, PAYMENT_CHANNEL_MISMATCH, URGENCY_PRESSURE

Example 2: a slip does not prove settlement

INPUT / Tuition slip present, no matching fictional ledger entry

RESULT / REQUIRES_OFFICIAL_CHECK

Flag: SLIP_NOT_FOUND_IN_SYNTHETIC_LEDGER | BANK SETTLEMENT: NOT VERIFIED

Institutional buyer is a hypothesis

TARGET USER

Students reviewing high-stakes tuition or deposit requests, with English and Indonesian sample communications.

BUYER TO VALIDATE

University international offices or student-support programs. No contract, pilot, payment or willingness-to-pay evidence.

Measure false clearance, safe action completion, user comprehension and time vs manual campus-policy check.

Finish the agent before final submission

CREATED NOW

Draft technical documentation; draft security assessment; editable deck; actual offline baseline logs and tests.

STILL MISSING

Model-enabled agent; real authentication; sandbox action audit; final evidence logs; public judge repo; actual screen-demo video.

No submission or real-world security claim is authorized by these artifacts.